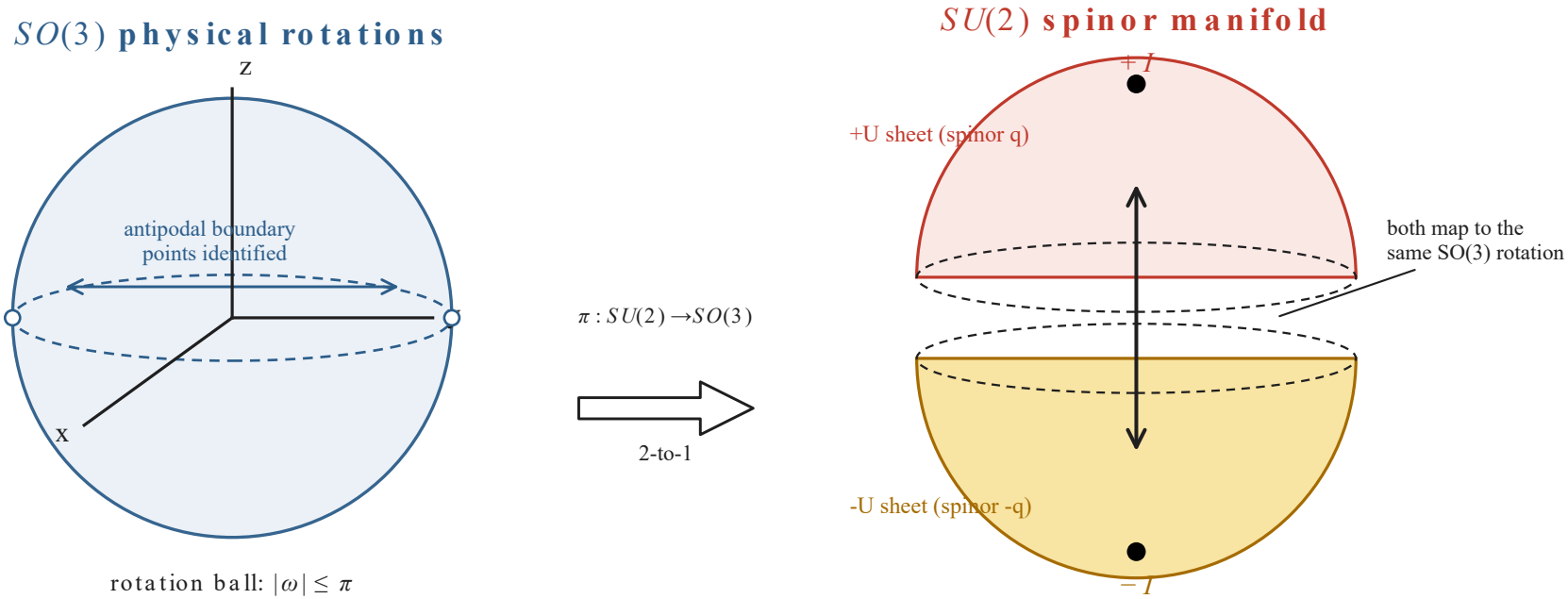
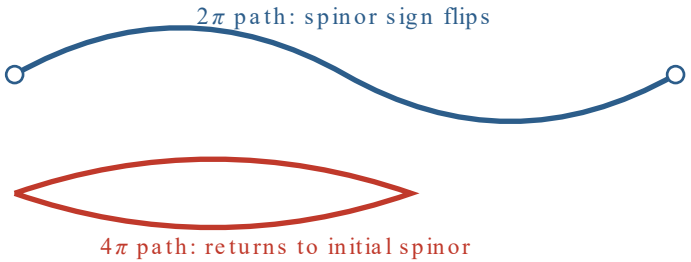
SU(2): The Double Cover of Spatial Rotations



Representative paths



Topological readout

SU(2) contains two spinor states, q and -q,
for each physical SO(3) rotation.
A single full turn is not homotopic to the identity;
two full turns are.

Fermions are SU(2) representations: spin-1/2 follows from the double cover.